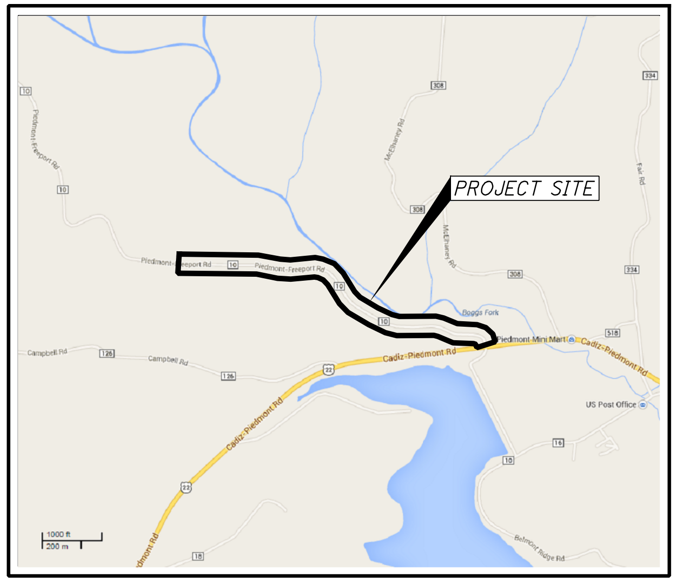
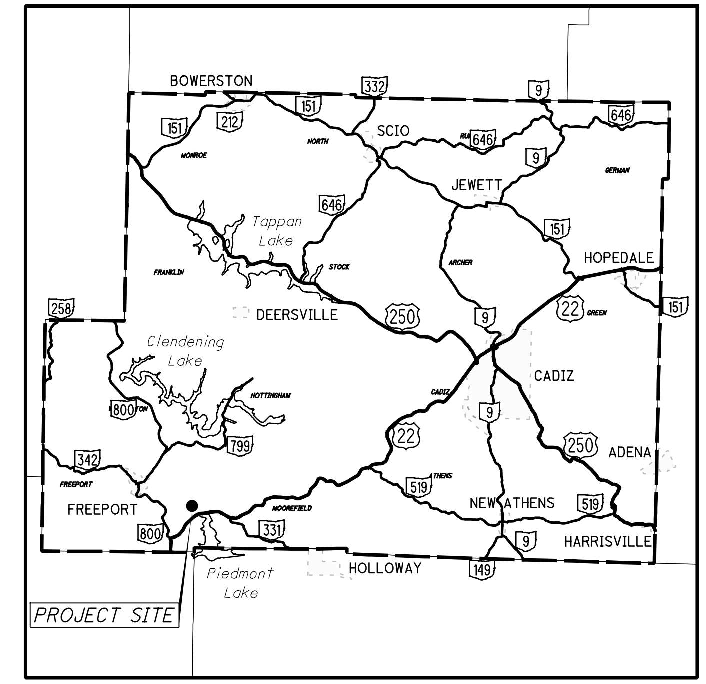
AMERICAN ENERGY UTICA
WELL SITE:
BOUSKA

FREEPORT TOWNSHIP
MOOREFIELD TOWNSHIP
HARRISON COUNTY
OHIO

PIEDMONT-FREEPORT RD. (C.R. 10) - 1.19 MI.

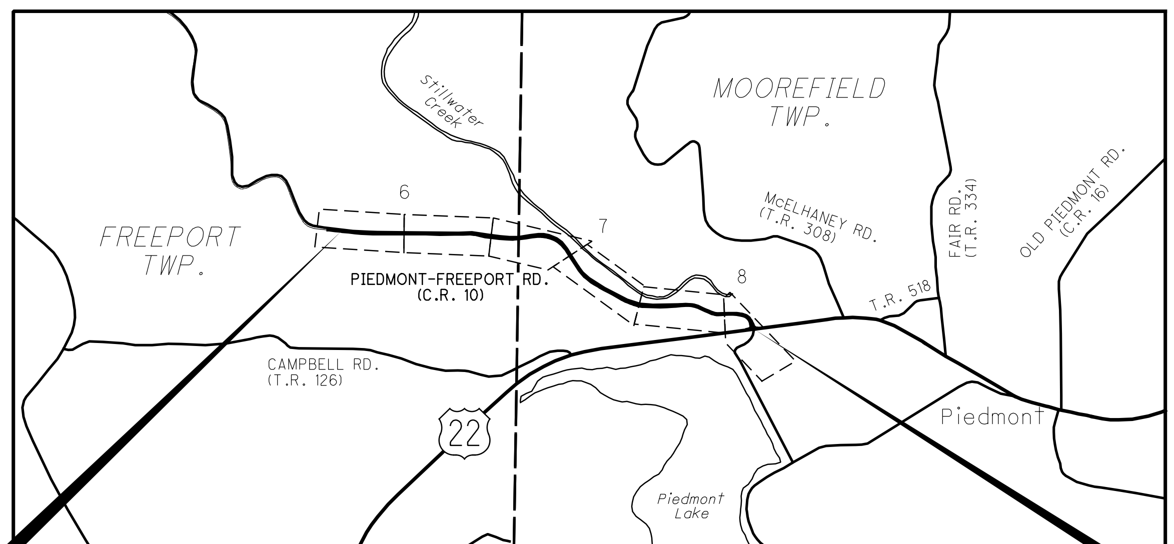


VICINITY MAP



LOCATION MAP
HARRISON COUNTY, OHIO

INDEX OF SHEETS:



BEGIN PROJECT
C.R. 10, STA. 110+00.00
40.196920,-81.234223

PLAN INDEX
SCALE IN MILES
0 1/4 1/2 1

END PROJECT
C.R. 10, STA. 173+33.64
40.192939,-81.213390

PROJECT DESCRIPTION

THIS PROJECT CONSISTS OF THE FOLLOWING WORK: FULL DEPTH RECLAMATION, PAVEMENT OVERLAY, WIDENING, AND INTERSECTION IMPROVEMENTS.

2013 SPECIFICATIONS

THE STANDARD SPECIFICATIONS OF THE STATE OF OHIO, DEPARTMENT OF TRANSPORTATION, INCLUDING CHANGES AND SUPPLEMENTAL SPECIFICATIONS LISTED IN THE PROPOSAL SHALL GOVERN THIS IMPROVEMENT.

CONVENTIONAL SIGNS

STATE LINE
COUNTY LINE
CORPORATION LINE
R/W PROPOSED R/W & EASEMENT LINE
Ex R/W EXISTING R/W LINE
PROPERTY LINE
EXISTING FENCE
PROPOSED FENCE
EDGE OF STREAM
PROPOSED GUARD RAIL
EXISTING GUARD RAIL
RAILROAD
G GAS LINE
W WATER LINE
T TELEPHONE LINE
E ELECTRIC LINE
DITCH CHECKS
FILTER SOCK
TELEPHONE POLE
POWER POLE
COMBINED POWER AND TELEPHONE POLE
TREE
SHRUB

APPROVED ______________________
DATE __________ AMERICAN ENERGY-UTICA, LLC

APPROVED ______________________
DATE __________ E.L. ROBINSON ENGINEERING

OHIO DEPARTMENT OF TRANSPORTATION STANDARD CONSTRUCTION DRAWINGS							
BP-3.1	7/18/14	MT-97.11	7/18/14				
		MT-99.20	7/19/13				
MGS-1.1	7/19/13						
MGS-2.1	7/19/13	TC-41.20	10/18/13				
MGS-4.1	7/19/13	TC-42.20	10/18/13				

UNDERGROUND UTILITIES
CONTACT BOTH SERVICES
CALL TWO WORKING DAYS
BEFORE YOU DIG
CALL
1-800-362-2764
(TOLL FREE)
OHIO UTILITIES PROTECTION SERVICE
NON-MEMBERS
MUST BE CALLED DIRECTLY
OIL & GAS PRODUCERS UNDERGROUND PROTECTION SERVICE CALL: **1-800-925-0988**

REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
TITLE SHEET
PIEDMONT-FREEPORT RD. (C.R. 10)
FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

PLANS PREPARED BY:
E.L. ROBINSON ENGINEERING
1801 Watermark Drive, Suite 310 · Columbus, Ohio 43215
www.elrobinsonengineering.com

BOUSKA WELL

NOTE	ITEM #	DESCRIPTION	PIEDMONT-FREEPORT RD. (C.R. 10) QUANTITY	UNIT
		ROADWAY		
	201	CLEARING AND GRUBBING	1	LUMP
	203	EXCAVATION, AS PER PLAN	800	CU YD
	206	CEMENT	420	TON
	SPECIAL	FULL DEPTH RECLAIMED BASE COURSE, 12 INCHES DEEP (6% TYPE 1 CEMENT, SOIL WT=110 PCF; SPREAD RATE = 65 PSY)	14,500	SQ YD
1	606	GUARDRAIL REBUILT, TYPE 5, AS PER PLAN	2,200	FT
		DRAINAGE		
	611	18" CONDUIT, TYPE B, HDPE	70	FT
		EROSION CONTROL		
	659	SEEDING AND MULCHING	151	SQ YD
		PAVEMENT		
	448	ASPHALT CONCRETE INTERMEDIATE COURSE, TYPE 2, PG64-22	1,400	TON
	448	ASPHALT CONCRETE SURFACE COURSE, TYPE 1, PG64-22	1,000	TON
	617	COMPACTED AGGREGATE	260	TON
		TRAFFIC CONTROL		
	642	EDGE LINE, 6" WHITE, TYPE 1	2.38	MILE
	642	CENTER LINE, TYPE 1	1.19	MILE
		MAINTENANCE OF TRAFFIC		
	614	MAINTAINING TRAFFIC	1	LUMP
		GENERAL		
	624	MOBILIZATION	1	LUMP

NOTE

1 REMOVAL OF EXISTING ITEM SHALL BE INCIDENTAL TO THE REPLACEMENT.

REV. NO. | SHEET NO. | DATE | DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
ESTIMATED QUANTITIES
PIEDMONT-FREEPORT RD. (C.R. 10)
FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

PLANS PREPARED BY:
E.L. ROBINSON ENGINEERING
1801 Watermark Drive, Suite 310 - Columbus, Ohio 43215
www.elrobinsonengineering.com

GENERAL NOTES

2013 SPECIFICATIONS
THE STANDARD SPECIFICATIONS OF THE STATE OF OHIO, DEPARTMENT OF TRANSPORTATION SHALL GOVERN THIS IMPROVEMENT. PURCHASING INFORMATION AND COPIES FOR VIEWING AND DOWNLOADING FOR THE ODOT 2013 CONSTRUCTION AND MATERIALS SPECIFICATIONS ARE AVAILABLE AT:

http://www.dot.state.oh.us/Divisions/ConstructionMgt/OnlineDocs/Pages/2013-Online-Spec-Book.aspx

MAINTENANCE OF TRAFFIC
ALL TRAFFIC CONTROL DEVICES SHALL BE FURNISHED, ERECTED, MAINTAINED, AND REMOVED BY THE CONTRACTOR IN ACCORDANCE WITH THE OHIO MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES (CURRENT EDITION). COPIES ARE AVAILABLE FOR VIEWING AND DOWNLOAD AT:

http://www.dot.state.oh.us/Divisions/Engineering/Roadway/DesignStandards/traffic/OhioMUTCD/Pages/OMUTCD2012_current_default.aspx

AND IS ALSO AVAILABLE FROM:
THE OHIO DEPARTMENT OF TRANSPORTATION
BUREAU OF TRAFFIC
1960 WEST BROAD STREET
COLUMBUS. OHIO 43223

CONSTRUCTION OPERATIONS SHALL NOT BEGIN UNTIL ALL TRAFFIC CONTROL IS IN PLACE AND APPROVED BY THE ON-SITE INSPECTOR. THE CONSTRUCTION INSPECTOR SHALL APPROVE ALL TEMPORARY TRAFFIC CONTROL DEVICES FOR CONDITION AND LOCATION BEFORE THE CONTRACTOR WILL BE ALLOWED TO BEGIN WORK. IF THE CONTRACTOR DOES NOT COMPLY WITH THE STANDARDS, ALL HIS WORK SHALL BE TERMINATED.

ALL WORK AND TRAFFIC CONTROL DEVICES SHALL BE IN ACCORDANCE WITH ODOT ITEM 614 AND OTHER APPLICABLE PORTIONS OF THE SPECIFICATIONS, AS WELL AS THE OHIO MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES. PAYMENT FOR ALL LABOR, EQUIPMENT, AND MATERIALS SHALL BE INCLUDED IN THE LUMP SUM CONTRACT PRICE FOR ITEM 614, MAINTAINING TRAFFIC, UNLESS SEPARATELY ITEMIZED IN THE PLAN.

IN THE EVENT THE CONTRACTOR FEELS A ROAD CLOSURE IS NEEDED TO COMPLETE THE WORK, THE ROAD CLOSURE SHALL BE APPROVED IN ADVANCE OF CONSTRUCTION COMMENCEMENT. THE REQUEST OF A ROAD CLOSURE DOES NOT GUARANTEE THE CLOSURE WILL BE PERMITTED UNLESS SPECIFICALLY NOTED IN THE PLANS.

CONTRACTOR SHALL HOLD A DAILY MEETING WITH THE ON-SITE INSPECTOR TO DISCUSS PROJECT SCOPE OF WORK AND PROPOSED MAINTENANCE OF TRAFFIC PLAN.

ALL ROAD CONSTRUCTION SIGNS INCLUDING ONE LANE ROAD SIGNS AND FLAGGER AHEAD SIGNS SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR. ALL TEMPORARY CONSTRUCTION SIGNS SHALL BE ON PORTABLE MOUNTS. FOLLOWING THE COMPLETION OF DAILY WORK THE CONTRACTOR SHALL REMOVE ALL TEMPORARY CONSTRUCTION SIGNS UNLESS CONDITIONS WARRANT OVERNIGHT SIGNAGE.

SIGN AND SUPPORT INSTALLATION SHALL BE IN ACCORDANCE WITH THE OHIO MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES.

WORK LIMITS
THE WORK LIMITS SHOWN ON THESE PLANS ARE FOR PHYSICAL CONSTRUCTION ONLY. PROVIDE THE INSTALLATION AND OPERATION OF ALL WORK ZONE TRAFFIC CONTROL AND WORK ZONE TRAFFIC CONTROL DEVICES REQUIRED BY THESE PLANS WHETHER INSIDE OR OUTSIDE THESE WORK LIMITS.

PROTECTION OF RIGHT-OF-WAY LANDSCAPING
CONTRACTOR SHALL BE RESPONSIBLE FOR REPAIRING ANY DAMAGE TO EXISTING FEATURES SCHEDULED TO REMAIN OR TO PRIVATE PROPERTY OUTSIDE OF CONSTRUCTION LIMITS.

UNDERGROUND FACILITIES
THE CONTRACTOR SHALL GIVE NOTICE OF INTENT TO CONSTRUCT TO THE OHIO UTILITIES PROTECTION SERVICE, THE OIL & GAS PRODUCERS UNDERGROUND PROTECTION SERVICE, AND OWNERS OF UNDERGROUND FACILITIES THAT ARE NOT MEMBERS OF A REGISTERED PROTECTION SERVICE IN ACCORDANCE WITH SECTIONS 153.64 AND 3781.25 TO 3781.32 OF THE OHIO REVISED CODE. THE ABOVE MENTIONED NOTICE SHALL BE GIVEN AT LEAST TWO WORKING DAYS PRIOR TO THE START OF CONSTRUCTION.

OHIO UTILITY PROTECTION SERVICE (OUPS)
1-800-362-2764

OIL & GAS PRODUCERS UNDERGROUND PROTECTION SERVICE (OGPUPS)
1-614-715-2984

NON MEMBERS MUST BE CALLED DIRECTLY.

EXISTING UNDERGROUND GAS LINES
CONTRACTOR SHALL VERIFY THE DEPTHS OF UNDERGROUND GAS LINE CROSSINGS THROUGH THE WORK AREA. IF THE DEPTH OF A LINE IS IN CONFLICT WITH THE PROPOSED WORK, CONTRACTOR SHALL COORDINATE LOWERING OF THE CONFLICTING LINE WITH THE UTILITY OWNER.

CLEAN-UP
TEMPORARY SOIL EROSION AND SEDIMENT CONTROL DEVICES SHALL BE REMOVED UPON COMPLETION OF EARTHWORK.

EXISTING CONDITIONS
THE EXISTING CONDITIONS HAVE BEEN APPROXIMATED USING READILY AVAILABLE AERIAL IMAGES AND LIDAR DATA AND LIMITED FIELD SURVEY INFORMATION. A SAMPLE OF EXISTING PAVEMENT WIDTHS AND THE PRESENCE OF EXISTING UTILITIES WERE COLLECTED. PAVEMENT WIDTHS AND QUANTITIES SHOWN IN THE PLANS ARE BASED ON THE AVERAGE PAVEMENT WIDTH. THE ENGINEER MAKES NO GUARANTEE OF THE ACCURACY OR COMPLETENESS OF THE BASE INFORMATION. EXISTING ROADS, DRIVEWAYS, STRUCTURES, AND UTILITIES SHOWN SHALL BE VERIFIED BY THE CONTRACTOR PRIOR TO CONSTRUCTION.

NECESSARY PRECAUTIONS SHALL BE TAKEN BY THE CONTRACTOR TO LOCATE AND PROTECT EXISTING UTILITY SERVICES AND MAINS. ANY DISCREPANCIES BETWEEN THE PLANS AND ACTUAL FIELD CONDITIONS SHALL BE REPORTED TO THE FIELD ENGINEER PRIOR TO CONSTRUCTION.

CONSTRUCTION
FAILURE TO SPECIFICALLY MENTION ANY WORK WHICH WOULD NORMALLY BE REQUIRED TO COMPLETE THE PROJECT SHALL NOT RELIEVE THE CONTRACTOR OF HIS RESPONSIBILITY TO PERFORM SUCH WORK. THE CONTRACTOR IS SOLELY RESPONSIBLE FOR THE CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES, PROCEDURES, AND SAFETY PRECAUTIONS AND PROGRAMS. THE CONTRACTOR IS RESPONSIBLE FOR TRANSITIONS TO DRIVEWAYS, DRAINAGE STRUCTURES, AND OTHER ROADSIDE FEATURES. METAL TRENCH PLATES ARE REQUIRED FOR ALL JOBS INVOLVING CULVERT REPLACEMENT. TRENCH PLATES MUST BE PRESENT ON SITE AND READY FOR IMMEDIATE USE PRIOR TO ROADWAY EXCAVATION.

DRIVE TIE-INS
EXISTING DRIVEWAYS SHALL BE TIED IN AT THE END OF EACH WORK DAY. DRIVEWAYS SHALL BE SLOPED BACK FROM THE EDGE OF ROADWAY AT A MAXIMUM GRADE OF 5%.

EXISTING TURF DRIVEWAYS SHALL BE SLOPED BACK FROM THE EDGE OF ASPHALT USING ITEM 304.

EXISTING GRAVEL DRIVEWAYS SHALL BE SLOPED BACK FROM THE EDGE OF ASPHALT USING ITEM 304.

EXISTING ASPHALT DRIVEWAYS SHALL BE SLOPED BACK FROM THE EDGE OF ASPHALT USING SPECIFIED ASPHALT SURFACE COURSE.

EXISTING CONCRETE DRIVEWAYS SHALL BE REPLACED IN-KIND.

ASPHALT WEDGES SHALL BE PLACED WHERE NOTED. THIS WORK SHALL CONSIST OF WINGING OUT THE PAVER ONE FOOT ACROSS THE DRIVEWAY.

ITEM 203, EXCAVATION, AS PER PLAN
EXCAVATION, AS PER PLAN INCLUDES THE EXCAVATION AND REMOVAL OF ANY AND ALL MATERIALS ENCOUNTERED, INCLUDING EXISTING PAVEMENT, NECESSARY FOR THE CONSTRUCTION OF THE PROPOSED WORK.

WASTE/BORROW SITES
CONTRACTOR SHALL BE RESPONSIBLE FOR FINDING OWN MATERIAL DISPOSAL SITE/BORROW SITE. ALL SITES MUST BE SUBMITTED FOR APPROVAL PRIOR TO USE. CONTRACTOR WILL BE RESPONSIBLE FOR OBTAINING A SIGNED AGREEMENT WITH THE LANDOWNER, PERFORMING NECESSARY REVIEWS, AND FOR DESIGNS AND FOR CLEANUP AND SEEDING.

WATERING
PERFORM DUST CONTROL OPERATIONS AND MAINTAIN CONTROL OF THE APPLICATION OF WATER AT ALL TIMES TO MINIMIZE DUST BUT NOT TO CREATE SATURATED SOIL CONDITIONS. FURNISH AND APPLY WATER USED FOR DUST CONTROL BY MEANS OF TANKS EQUIPPED WITH SUITABLE SPRINKLING DEVICES.

QUANTITIES
CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING ALL BID QUANTITIES AND ITEMS OF WORK PRIOR TO SUBMITTING PROJECT BID.

PERMITS
CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS TO COMPLETE THE WORK.

DETOUR ROUTE
CONTRACTOR SHALL BE RESPONSIBLE FOR ALL DETOUR AND ROAD CLOSURE SIGNAGE, IMPLEMENTATION, AND COORDINATION OF THE DETOUR ROUTE WITH HARRISON COUNTY, VILLAGE OF DEERSVILLE, FRANKLIN TOWNSHIP, AND STOCK TOWNSHIP AUTHORITIES.

PROFILE AND ALIGNMENT
PLACE THE PROPOSED PAVEMENT TO FOLLOW THE ALIGNMENT AND PROFILE OF THE EXISTING PAVEMENT AND AS SHOWN ON THE TYPICAL SECTIONS.

MONUMENT BOXES
CONTRACTOR SHALL ADJUST MONUMENT BOXES TO GRADE PER ODOT 604. PAYMENT FOR THIS ITEM SHALL BE INCIDENTAL TO THE PROJECT.

STREET SWEEPING
CONTRACTOR IS RESPONSIBLE FOR MAINTAINING THE ROUTE AT THE ENTRANCE TO THE HAUL ROUTE IN A SAFE CONDITION THAT IS FREE FROM MUD AND DEBRIS. CONTRACTOR SHALL HAVE A STREET SWEEPER AVAILABLE AT ALL TIMES TO MEET THIS REQUIREMENT.

ITEM 614, MAINTAINING TRAFFIC
ACCESS TO ALL DRIVES SHALL BE MAINTAINED AT ALL TIMES DURING CONSTRUCTION.

ACCESS FOR MAIL DELIVERY, EMERGENCY, AND SERVICE VEHICLES SHALL NOT BE DISRUPTED. THE METHOD TO PROVIDE THIS ACCESS IS TO BE DETERMINED BY THE CONTRACTOR; HOWEVER, THE METHOD(S) MUST BE IN ACCORDANCE WITH CMS ITEM 614 AND THE OHIO MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES AND BE APPROVED BY THE COUNTY ENGINEER'S OFFICE. THE CONTRACTOR SHALL COORDINATE CONSTRUCTION ACTIVITIES WITH THE ENGINEER AND THE OWNERS OF THE ABUTTING PROPERTIES IN ADVANCE (10 DAYS) OF ANY OPERATIONS WHICH AFFECT ACCESS. PAYMENT FOR ALL LABOR, MATERIALS, AND EQUIPMENT REQUIRED TO PROVIDE THE ABOVE SERVICES SHALL BE INCLUDED IN THE PRICE BID FOR ITEM 614, MAINTAINING TRAFFIC.

CONTRACTOR DISTURBED AREAS
THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING SEEDING AND FILTER SOCKS FOR ALL AREAS BEYOND THE PAVEMENT LIMITS THAT ARE DISTURBED BY THE CONTRACTOR. THESE MEASURES SHALL BE IN ACCORDANCE WITH THE EROSION CONTROL DETAILS PROVIDED IN THIS PLAN.

PROJECT NOTES

PROJECT CONTACTS
AMERICAN ENERGY-UTICA
JEFF BECK, FIELD SUPERINTENDENT
1000 UTICA WAY
CAMBRIDGE, OH 43725
740-432-9020

E.L. ROBINSON ENGINEERING
BRENT DOWNING, P.E.
1801 WATERMARK DRIVE, SUITE 310
COLUMBUS, OH 43215
614-586-0642

HARRISON COUNTY ENGINEER
ROBERT STERLING, P.E., P.S.
100 WEST MARKET ST
COURT HOUSE
CADIZ, OH 43907
740-942-8867

TACK COAT
PAYMENT FOR ITEM 407 TACK COAT, MATERIALS AND COST TO INSTALL SHALL BE INCIDENTAL TO ITEM 448 ASPHALT CONCRETE SURFACE COURSE.

REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
GENERAL NOTES
PIEDMONT-FREEPORT RD. (C.R. 10)
FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

PLANS PREPARED BY:
E.L. ROBINSON ENGINEERING
1801 Watermark Drive, Suite 310 · Columbus, Ohio 43215
www.elrobinsonengineering.com

PROJECT NOTES (CONT'D.)

PAVEMENT CROSS SLOPES

CONTRACTOR SHALL VERIFY THE EXISTING PAVEMENT CROSS SLOPES. CONTRACTOR SHALL ENSURE THE FINAL PAVEMENT SURFACE DOES NOT CONTAIN LOCALIZED DEPRESSIONS AND IS SLOPED TO DRAIN.

PAVEMENT MARKINGS

CONTRACTOR SHALL VERIFY THE EXISTING PAVEMENT MARKINGS AND MATCH THE EXISTING PAVEMENT MARKINGS AND LOCATIONS FOR ALL WORK AREAS OR AS SHOWN IN THE PLANS.

REGULATORY SIGNS

IN THE EVENT A REGULATORY SIGN NEEDS TO BE RELOCATED, THE FINAL SIGN PLACEMENT SHALL BE APPROVED BY THE REGULATORY AGENCY.

TREE TRIMMING

CONTRACTOR SHALL BE RESPONSIBLE FOR TRIMMING TREES WITHIN THE LIMITS OF ROADWAY DITCH LINES TO A HEIGHT OF 17 FEET FROM THE PAVEMENT SURFACE.

ITEM 606, GUARDRAIL REBUILT, TYPE 5, AS PER PLAN

THIS WORK SHALL INCLUDE ALL MATERIALS, LABOR AND INCIDENTALS REQUIRED TO REBUILD THE EXISTING GUARDRAIL INCLUDING THE FOLLOWING:

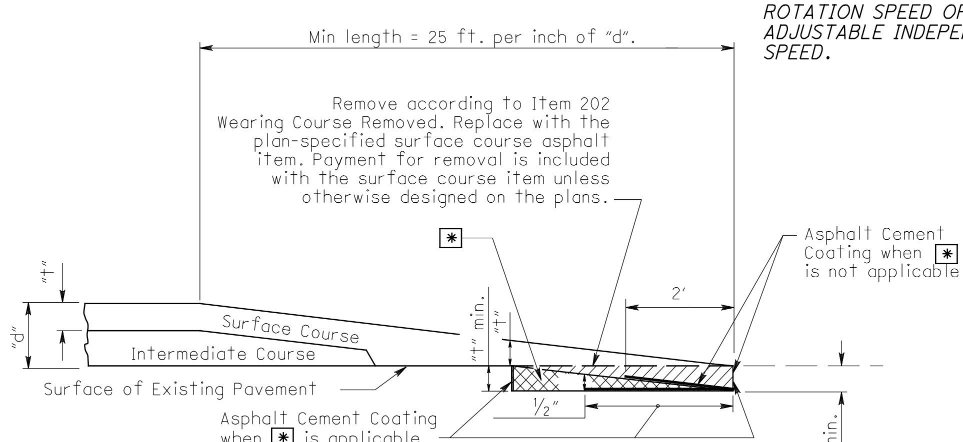
1. REBUILDING GUARDRAIL TO HEIGHT PER ODOT STANDARD CONSTRUCTION DRAWING GR-1.1.

2. REBUILDING ANCHOR ASSEMBLIES PER ODOT ITEM 606, ANCHOR ASSEMBLY REBUILT, TYPE A.

3. INSTALLING ADDITIONAL BEAMS AND ASSOCIATED AGGREGATE TO SUPPLEMENT THE EXISTING RETAINING STRUCTURE IN THE VICINITY OF STA. 161+50.

PAVEMENT FEATHERED AREAS

PROVIDE BUTT JOINT TYPE AT INTERSECTIONS PER ODOT STD. DWG. BP-3.1.



BUTT JOINT TYPE

✱ Permissible removal and replacement.

SPECIAL, FULL DEPTH RECLAIMED BASE COURSE, 12 INCHES DEEP

1. DESCRIPTION

FULL DEPTH RECLAMATION (FDR) CONSISTS OF CONSTRUCTING A STABILIZED, RECLAIMED BASE COURSE BY PULVERIZING THE EXISTING ASPHALT PAVEMENT AND EXISTING AGGREGATE MATERIAL, AND THEN MIXING IT WITH THE SUBGRADE SOIL AND EITHER TYPE 1 PORTLAND CEMENT.

CONTRACTOR SHALL PREGRIND AND PROFILE TO MATCH EXISTING CROSS SLOPES AND PROVIDE A MINIMUM CROSS SLOPE OF 2% FOR NORMAL CROWN ROADWAYS.

2. MATERIALS

FURNISH MATERIALS CONFORMING TO ODOT C&MS FOR:
- PORTLAND CEMENT, 701.04
- AGGREGATE, 703.04

FURNISH WATER CONFORMING TO 499.02.

FOR THE CURING COAT, FURNISH RAPID SETTING EMULSIFIED ASPHALT CONFORMING TO 702.04 OR THE CURING MATERIALS SPECIFIED IN 451.02.

3. SUBMITTALS

PREPARE AND SUBMIT THE REPORT FOR MIXTURE DESIGN FOR FULL DEPTH RECLAMATION AS REQUIRED IN SUPPLEMENT 1120 AND MODIFIED IN THIS SPECIAL PROVISION.

SUBMIT, FOR THE ENGINEER'S ACCEPTANCE, A REPORT THAT LISTS THE TYPE OF EQUIPMENT TO BE USED, SPEED OF THE INTENDED EQUIPMENT USAGE, RATE OF APPLICATION OF THE CHEMICAL, AND CALCULATIONS THAT DEMONSTRATE HOW THE REQUIRED PERCENTAGE OF CHEMICAL WILL BE APPLIED. SUBMIT THE REPORT TO THE ENGINEER FOR ACCEPTANCE AT LEAST 2 WORKDAYS BEFORE THE STABILIZATION WORK BEGINS.

4. EQUIPMENT

PROVIDE EQUIPMENT THAT MEETS THE FOLLOWING REQUIREMENTS:

A. USE EQUIPMENT CAPABLE OF AUTOMATICALLY METERING LIQUIDS WITH A VARIATION OF NOT MORE THAN TWO PERCENT BY WEIGHT OF LIQUIDS. CALIBRATE BEFORE USE.

B. RECLAIMER. USE A SELF-PROPELLED, TRAVELING ROTARY RECLAIMER OR EQUIVALENT MACHINE CAPABLE OF CUTTING THROUGH EXISTING ROADWAY MATERIAL TO DEPTHS OF UP TO 16 INCHES (405 MM) WITH ONE PASS. PROVIDE EQUIPMENT CAPABLE OF PULVERIZING THE EXISTING PAVEMENT, AGGREGATE BASE AND SUBGRADE SOIL IN PLACE, TO A MINIMUM WIDTH OF 8 FEET (2.4 METERS), AND MIXING ANY ADDED MATERIALS TO THE SPECIFIED DEPTH. THE ROTATION SPEED OF THE CUTTING DRUM MUST BE ADJUSTABLE INDEPENDENT OF THE MACHINE'S FORWARD SPEED.

USE A MACHINE EQUIPPED WITH A COMPUTER CONTROLLED LIQUID PROPORTIONING SYSTEM CAPABLE OF REGULATING AND MONITORING THE WATER APPLICATION RATE RELATIVE TO DEPTH OF CUT, WIDTH OF CUT, AND SPEED. CONNECT THE WATER PUMP ON THE MACHINE BY A HOSE TO THE SUPPLY TRUCK, AND MECHANICALLY OR ELECTRONICALLY INTERLOCK THE PUMP WITH THE FORWARD MOVEMENT OF THE MACHINE. MOUNT THE SPRAY BAR SO THAT WATER IS INJECTED DIRECTLY INTO THE MIXING CHAMBER. PROVIDE EQUIPMENT CAPABLE OF MIXING WATER, CHEMICALS, AND THE PULVERIZED PAVEMENT MATERIALS INTO A HOMOGENOUS MIXTURE. MAINTAIN THE CUTTING DRUM IN GOOD CONDITION DURING THE WORK.

DO NOT USE EQUIPMENT SUCH AS ROAD PLANERS OR COLD-MILLING MACHINES DESIGNED TO MILL OR PLANE THE EXISTING ROADWAY MATERIALS RATHER THAN CRUSH OR FRACTURE THEM.

C.COMPACTION. PROVIDE A VIBRATORY PADFOOT ROLLER WITH AT LEAST 52,000 POUNDS (23,500 KG) OF CENTRIFUGAL FORCE FOR BREAKDOWN COMPACTION. PROVIDE A SINGLE OR TANDEM SMOOTH DRUM VIBRATORY ROLLER HAVING A MINIMUM EFFECTIVE WEIGHT OF 12 TONS (11 METRIC TONS) FOR FINISH ROLLING.

5. CONSTRUCTION

PERFORM FULL DEPTH RECLAMATION WORK WHEN THE AIR TEMPERATURE IS 40°F (5°C) OR ABOVE AND WHEN THE SOIL IS NOT FROZEN. DO NOT PERFORM THIS WORK DURING WET OR UNSUITABLE WEATHER.

A. PULVERIZING AND SHAPING.
BEFORE SPREADING ANY STABILIZING CHEMICALS OR AGGREGATE, PULVERIZE THE EXISTING ROADWAY MATERIALS TO THE SPECIFIED DEPTH. PULVERIZE UNTIL 95 PERCENT OF THE MATERIAL PASSES THROUGH THE 2 INCH (50 MM) SIEVE. SHAPE TO WITHIN 3/4 INCH (18 MM) OF THE PROPOSED GRADE AND COMPACT UNTIL NO FURTHER DENSIFICATION IS ACHIEVED. ADD WATER TO THE PULVERIZED MATERIAL TO BRING IT TO AT LEAST OPTIMUM MOISTURE CONTENT BUT NOT MORE THAN 3 PERCENT ABOVE OPTIMUM MOISTURE CONTENT.

B. SPREADING.
AFTER PULVERIZING, SPREAD THE SPECIFIED CHEMICAL UNIFORMLY ON THE SURFACE USING A MECHANICAL SPREADER AT THE APPROVED RATE AND AT A CONSTANT SLOW RATE OF SPEED. USE A DISTRIBUTION BAR WITH A MAXIMUM HEIGHT OF 3 FEET (1 METER) ABOVE THE SUBGRADE. USE A CANVAS SHROUD THAT SURROUNDS THE DISTRIBUTION BAR AND EXTENDS TO THE SURFACE.

MINIMIZE DUSTING WHEN SPREADING THE CHEMICAL. CONTROL DUST ACCORDING TO 107.19. DO NOT SPREAD THE CHEMICAL WHEN WIND CONDITIONS CREATE BLOWING DUST THAT EXCEEDS THE LIMITS IN 107.19.

DO NOT SPREAD THE CHEMICAL ON STANDING WATER.

IF SPECIFIED, SPREAD AGGREGATE AT THE APPROVED RATE USING AN AUGER OR VANE TYPE DISTRIBUTOR FOR DRY MATERIALS.

PROVIDE A ONE SQUARE YARD (SQUARE METER) CANVAS SHEET AND SCALE TO CHECK THE SPREADING RATE OF THE CHEMICAL AND AGGREGATE.

C.MIXING.
MIX ACCORDING TO 206.05.B. MIX CEMENT OR LIME KILN DUST ACCORDING TO 206.05.B.1. MIX LIME ACCORDING TO 206.05.B.2. CHECK THE DEPTH OF THE MIXTURE ACCORDING TO 206.05.B.3.

D.COMPACTING.
START COMPACTION NO MORE THAN 30 MINUTES AFTER THE FINAL MIXING. BEGIN ROLLING AT THE LOW SIDE OF THE RECLAIMED BASE COURSE. INITIALLY, DO NOT COMPACT WITHIN 3 TO 6 INCHES (80 TO 150 MM) OF AN UNSUPPORTED EDGE TO PREVENT DISTORTION.

COMPACT THE RECLAIMED BASE COURSE TO THE REQUIREMENTS IN 204.03, EXCEPT THE ENGINEER WILL USE 98 PERCENT OF THE MAXIMUM DRY DENSITY FOR ACCEPTANCE. THE ENGINEER WILL DETERMINE THE MAXIMUM DRY DENSITY FOR ACCEPTANCE USING THE TEST SECTION METHOD.

USE THE MOISTURE CONTROLS ACCORDING TO 203.07.A, EXCEPT ENSURE THAT THE MOISTURE CONTENT AT TIME OF COMPACTION IS AT OR ABOVE OPTIMUM BUT NOT GREATER THAN 3 PERCENT ABOVE OPTIMUM MOISTURE CONTENT.

PERFORM THE FINAL ROLLING USING A STEEL-WHEELED ROLLER. DO NOT USE VIBRATION DURING THE FINAL ROLLING.

THE CONTRACTOR MAY EITHER SHAPE AND FINE GRADE THE RECLAIMED BASE COURSE BEFORE THE CURING PERIOD, OR SHAPE THE RECLAIMED BASE COURSE BEFORE THE CURING PERIOD AND FINE GRADE AFTER THE CURING PERIOD. IF FINE GRADING BEFORE THE CURING PERIOD, FINE GRADE THE SAME DAY AS MIXING, COMPACTING, AND SHAPING. IF FINE GRADING AFTER THE CURING PERIOD, SHAPE THE RECLAIMED BASE COURSE APPROXIMATELY 1/2 INCH (13 MM) ABOVE THE PROFILE GRADE AND TYPICAL SECTIONS. IN EITHER CASE, FINE GRADE THE RECLAIMED BASE COURSE TO THE PROFILE GRADE AND TYPICAL SECTIONS WITHIN THE TOLERANCES IN 203.08.

E.CURING.
IMMEDIATELY AFTER THE COMPACTION AND SHAPING OF THE RECLAIMED BASE COURSE, COVER THE SURFACE WITH CURING COAT FOR CURING THE RECLAIMED BASE COURSE. USE A RATE OF 1 GALLON PER 30 SQUARE FEET (1.36 LITERS PER SQUARE METER) FOR EMULSIONS OR A RATE OF 1 GALLON PER 150 SQUARE FEET (0.27 LITERS PER SQUARE METER) WHEN THE CURING MATERIALS IN 451.02 ARE USED.

APPLY THE CURING COAT PRIOR TO THE SURFACE DRYING OUT. IF THE CURING COAT IS DELAYED OR THE SURFACE STARTS TO DRY OUT APPLY WATER FOR TEMPORARY CURING UNTIL THE CURING COAT CAN BE APPLIED. DO NOT APPLY THE CURING COAT UNLESS THE CURING COAT CAN SET UP BEFORE IT RAINS. WHEN THE APPLICATION OF CURING COAT MUST BE DELAYED, KEEP THE RECLAIMED BASE COURSE WET BY USING WATER UNTIL THE CURING COAT CAN BE APPLIED. CURING COAT IS INCIDENTAL TO THE WORK.

CURE THE RECLAIMED BASE COURSE FOR AT LEAST FIVE DAYS BEFORE PLACEMENT OF THE OVERLYING COURSE.

F.PROOF ROLLING.
AFTER THE CURE PERIOD, PROOF ROLL THE RECLAIMED BASE COURSE ACCORDING TO ITEM 204.

G.PROTECTION.
PROTECT ANY FINISHED PORTION OF THE RECLAIMED BASE COURSE UPON WHICH ANY CONSTRUCTION EQUIPMENT IS REQUIRED TO TRAVEL TO PREVENT MARRING, DISTORTION OR DAMAGE OF ANY KIND. IMMEDIATELY AND SATISFACTORILY CORRECT ANY SUCH DAMAGE.

DRAIN AND MAINTAIN THE WORK ACCORDING TO 203.04.A. DO NOT OPERATE ANY EQUIPMENT ON THE RECLAIMED BASE COURSE DURING THE CURING PERIOD. DO NOT ALLOW THE RECLAIMED BASE COURSE TO FREEZE DURING THE CURE PERIOD. COVER THE COMPLETED RECLAIMED BASE COURSE WITH ASPHALT CONCRETE PAVEMENT WITHIN 14 CALENDAR DAYS.

REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR: AMERICAN ENERGY PARTNERS

BOUSKA WELL
GENERAL NOTES

PIEDMONT-FREEPORT RD. (C.R. 10)

FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

PLANS PREPARED BY:
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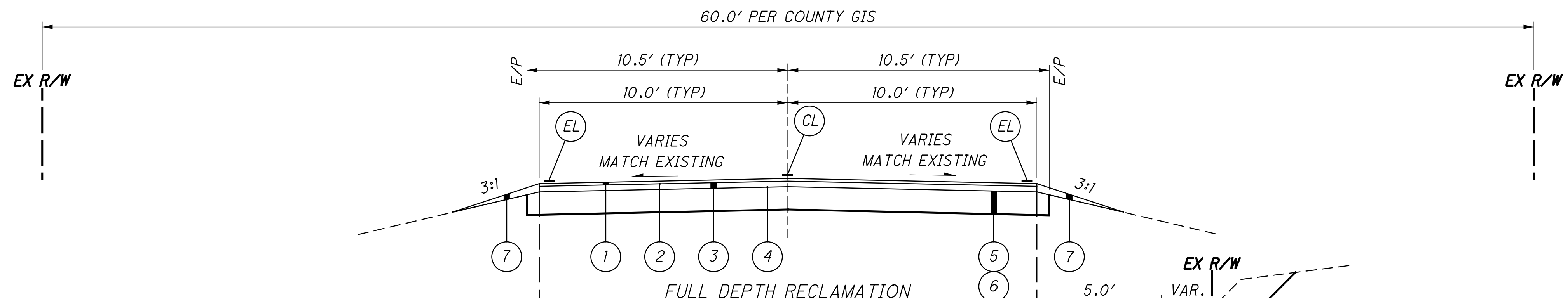
PROPOSED LEGEND

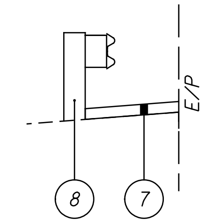
1. ITEM 448 - 1.25" ASPHALT CONCRETE SURFACE COURSE, TYPE 1, PG64-22
2. ITEM 407 - TACK COAT FOR INTERMEDIATE COURSE (0.04 GAL/SY)
3. ITEM 448 - 1.75" ASPHALT CONCRETE INTERMEDIATE COURSE, TYPE 2, PG64-22
4. ITEM 407 - TACK COAT (0.075 GAL/SY)
5. ITEM 206 - CEMENT
6. ITEM SPECIAL - FDR BASE COURSE, 12 INCHES DEEP (6% TYPE 1 CEMENT, SOIL WT=110 PCF; SPREAD RATE = 65 PSY)
7. ITEM 617 - COMPACTED AGGREGATE
8. ITEM 606 - GUARDRAIL REBUILT, TYPE 5
9. ITEM 659 - SEEDING & MULCHING
10. ITEM 203 - EXCAVATION, AS PER PLAN

(CL) ITEM 642 - CENTER LINE, TYPE 1

(EL) ITEM 642 - EDGE LINE, 6" WHITE, TYPE 1

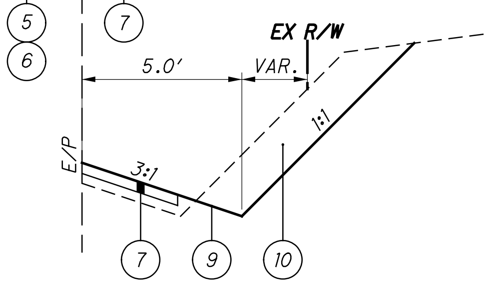


FULL DEPTH RECLAMATION
LIMITING STATIONS
C.R. 10 STA. 110+00.00 TO STA. 173+33.64



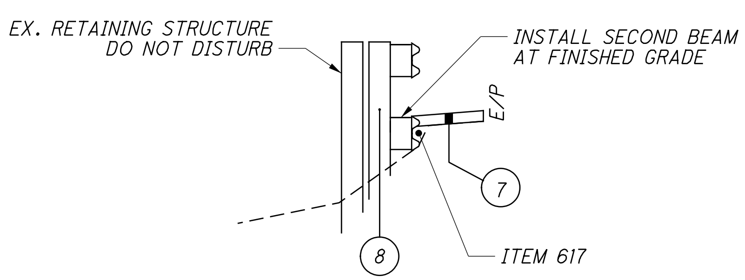
GUARDRAIL DETAIL

REPLACE AT EXISTING LOCATIONS.
REPLACE POSTS AND REUSE EXISTING RAIL.



DITCH DETAIL

STA. 170+94.80, RT TO STA. 173+44.19, RT



GUARDRAIL DETAIL

STA. 160+35 TO STA. 160+72.5
NOT TO SCALE

REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR: AMERICAN ENERGY PARTNERS

BOUSKA WELL
TYPICAL SECTION
PIEDMONT-FREEPORT RD. (C.R. 10)
FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

PLANS PREPARED BY:
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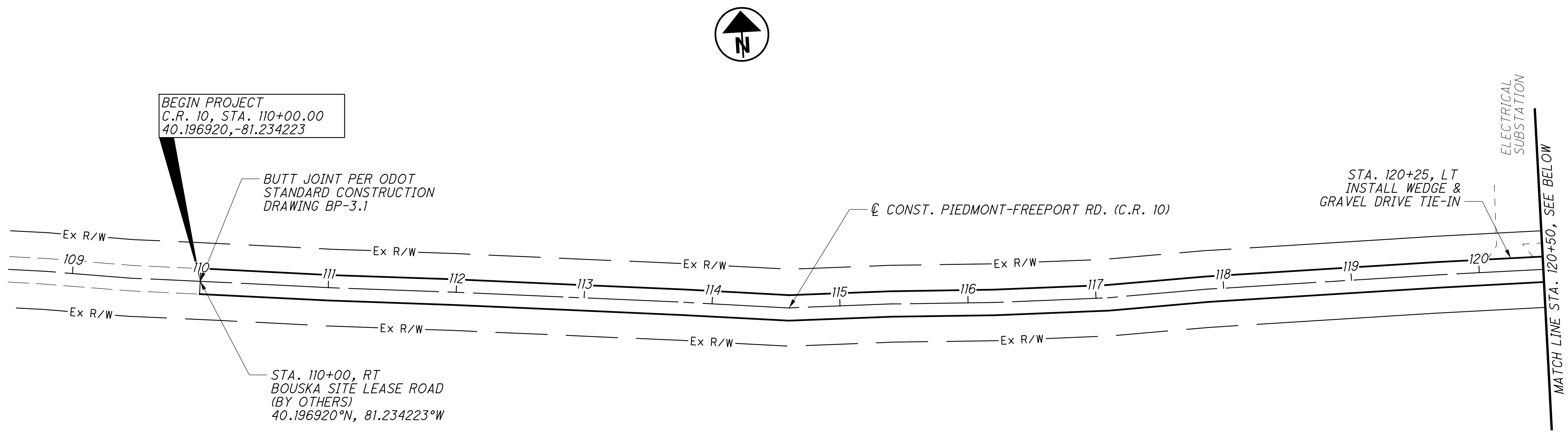
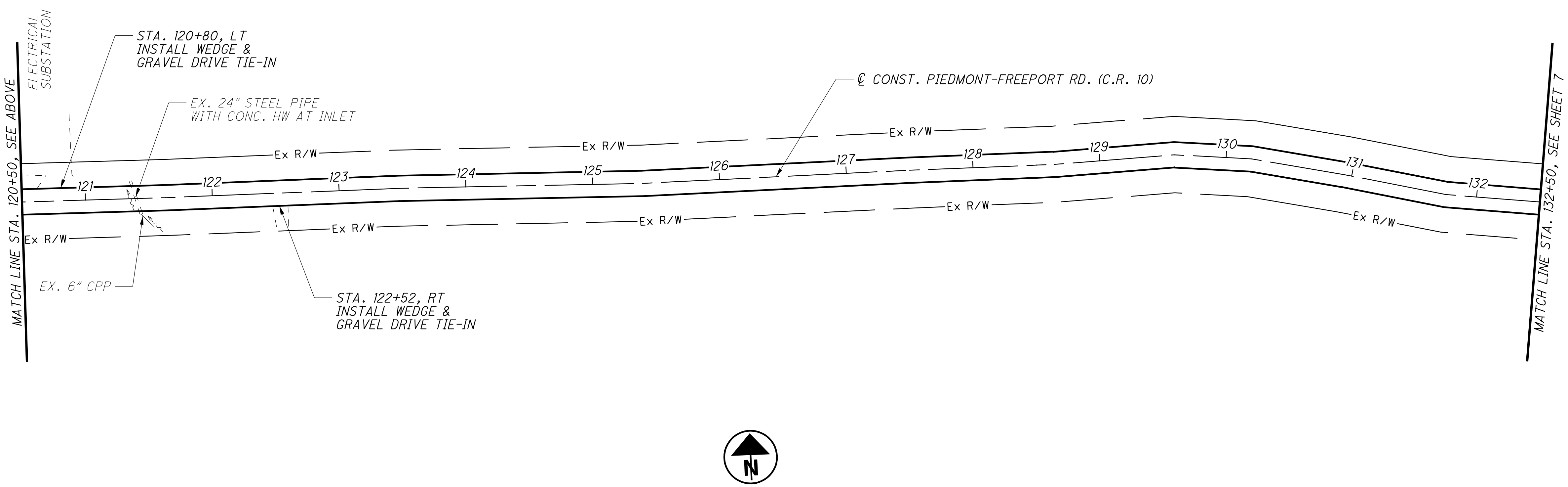


NOTE:
THE ROADWAY CENTERLINE, STATIONING, AND OFFSET DISTANCES ARE APPROXIMATE AND TO BE REFERENCED FOR GRAPHICAL PURPOSES ONLY.



REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
PLAN
PIEDMONT-FREEPORT RD. (C.R. 10) STA. 108+50 TO STA. 132+50

FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

0 25 50 100
HORIZONTAL SCALE IN FEET

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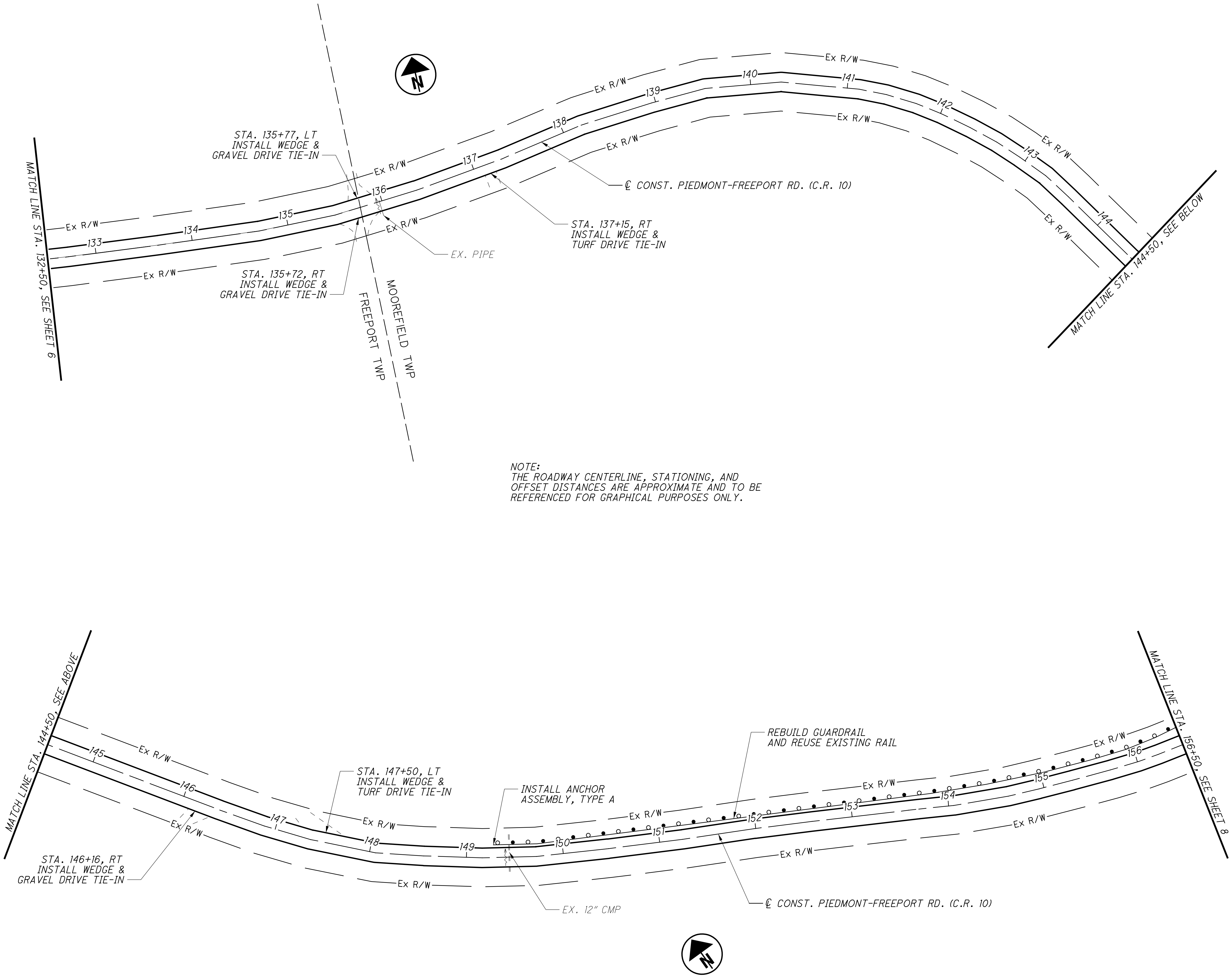

NOTE:
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REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
PLAN

PIEDMONT-FREEPORT RD. (C.R. 10)
STA. 132+50 TO STA. 156+50

FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

0 25 50 100
HORIZONTAL SCALE IN FEET

PLANS PREPARED BY:
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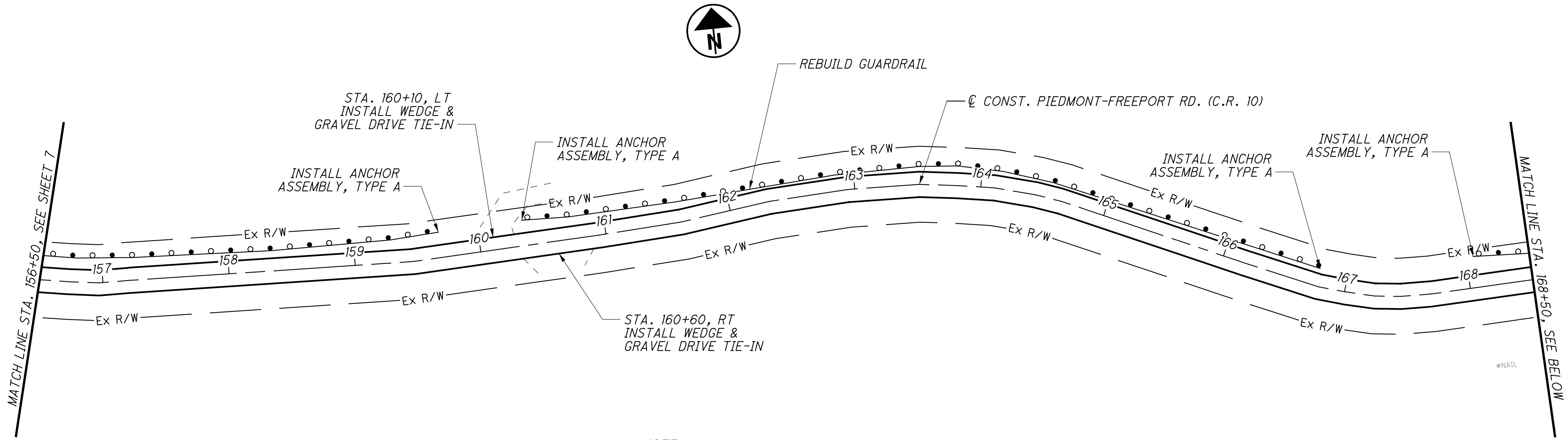

NOTE:
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OFFSET DISTANCES ARE APPROXIMATE AND TO BE
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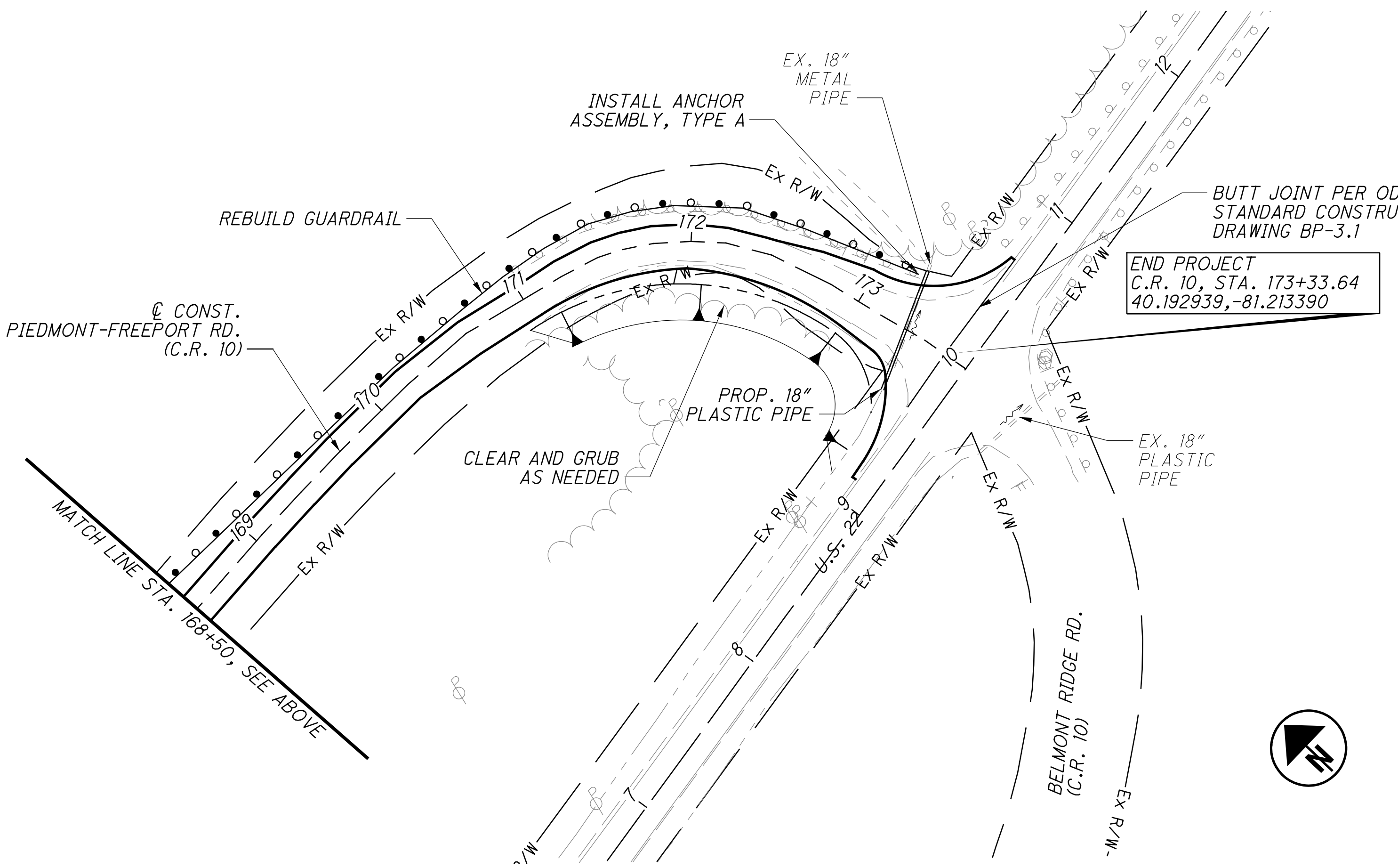
NOTE:
FOR INTERSECTION DETAILS, SEE SHEETS 9 & 10.



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PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
PLAN

PIEDMONT-FREEPORT RD. (C.R. 10)
STA. 156+50 TO STA. 173+56

FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

0 25 50 100
HORIZONTAL
SCALE IN FEET

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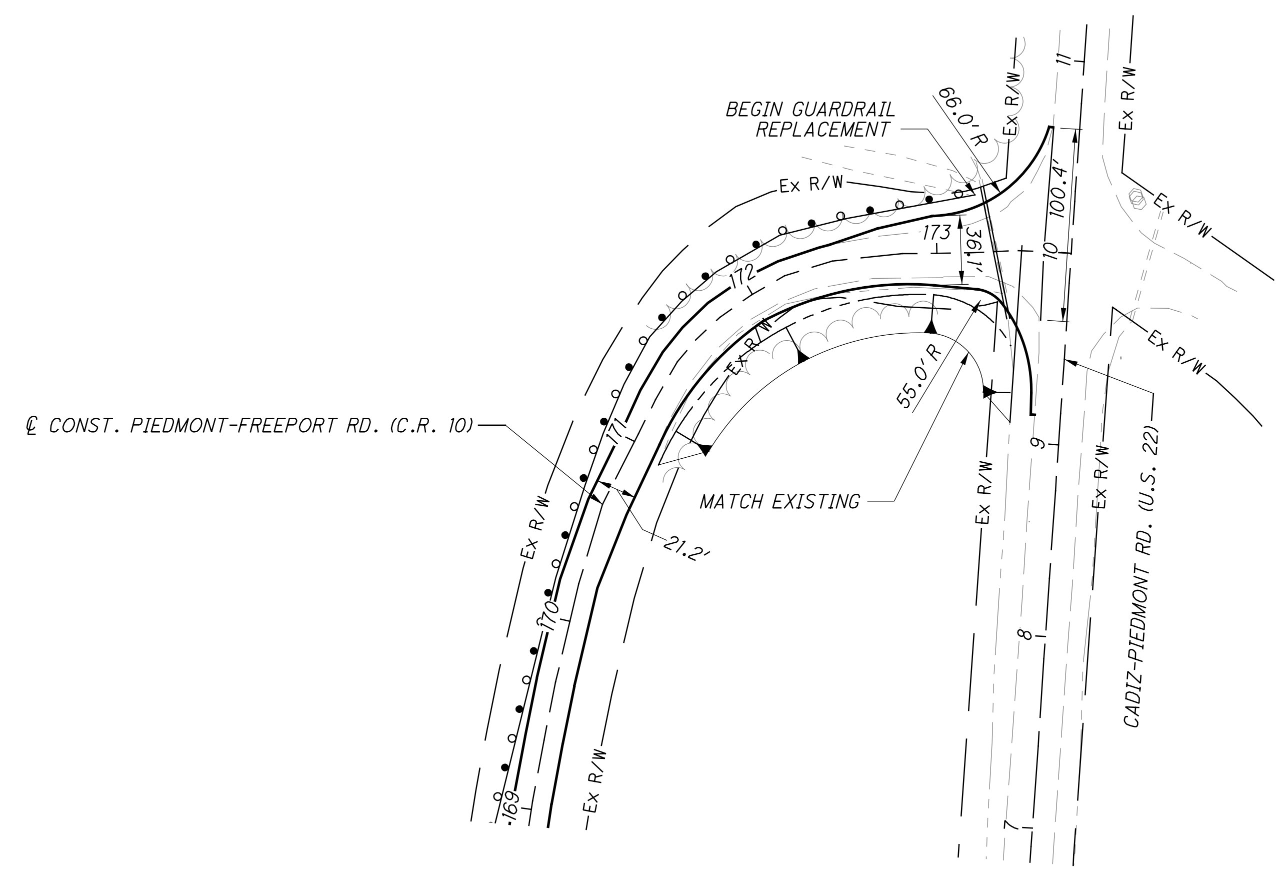

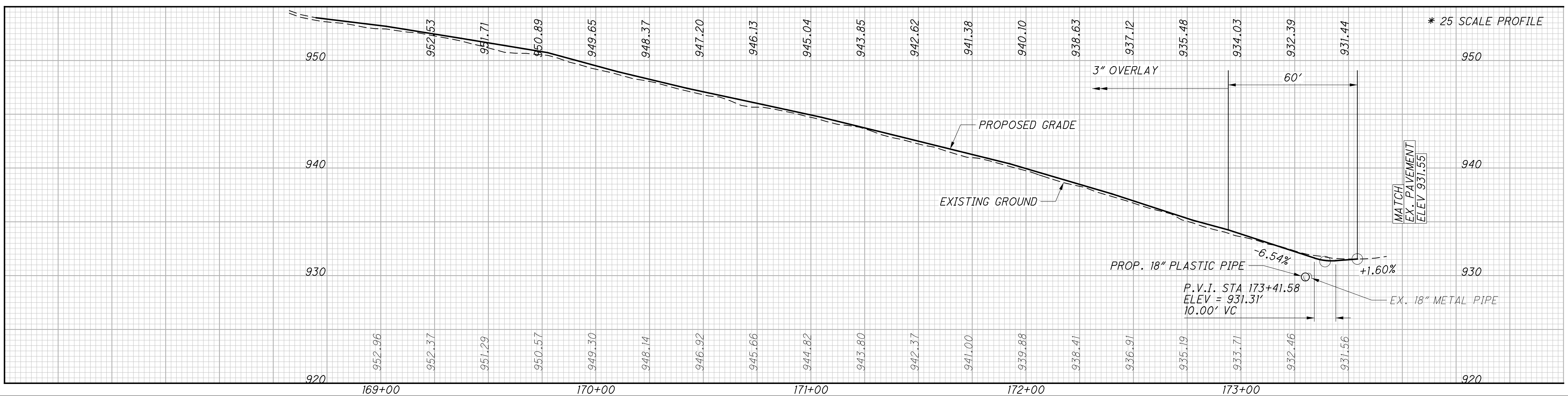

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PLANS PREPARED FOR:

AMERICAN ENERGY PARTNERS

BOUSKA WELL

PLAN AND PROFILE

PIEDMONT-FREEPORT RD. (C.R. 10)

FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

0 25 50 100
HORIZONTAL SCALE IN FEET

PLANS PREPARED BY:

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NOTE:
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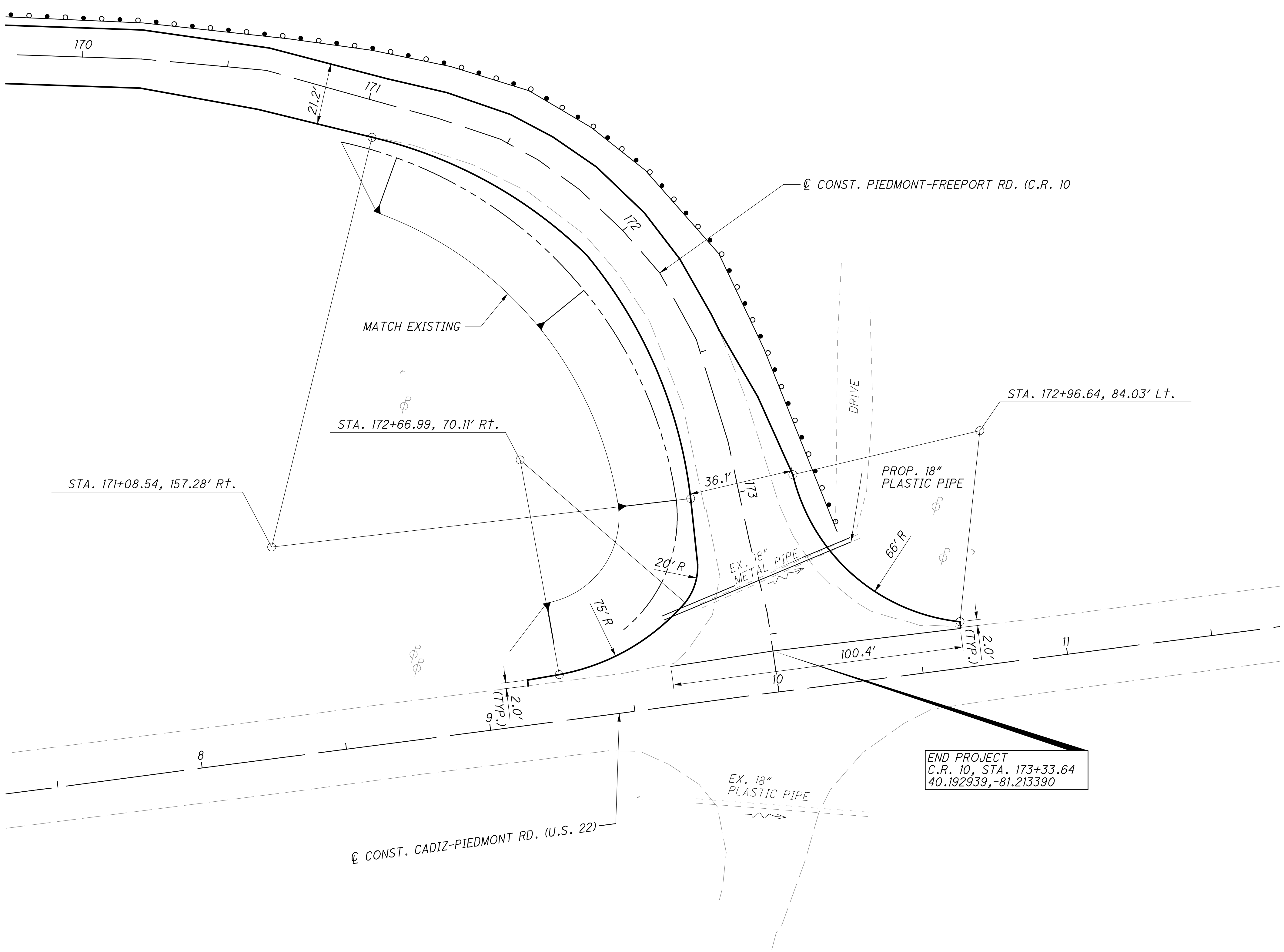

NOTE:
FOR TYPICAL SECTION, SEE SHEET 5

REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

BOUSKA WELL
INTERSECTION DETAILS
PIEDMONT-FREEPORT RD. (C.R. 10) & CADIZ-PIEDMONT RD. (U.S. 22)
FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

0 10 20 50
HORIZONTAL SCALE IN FEET

PLANS PREPARED BY:
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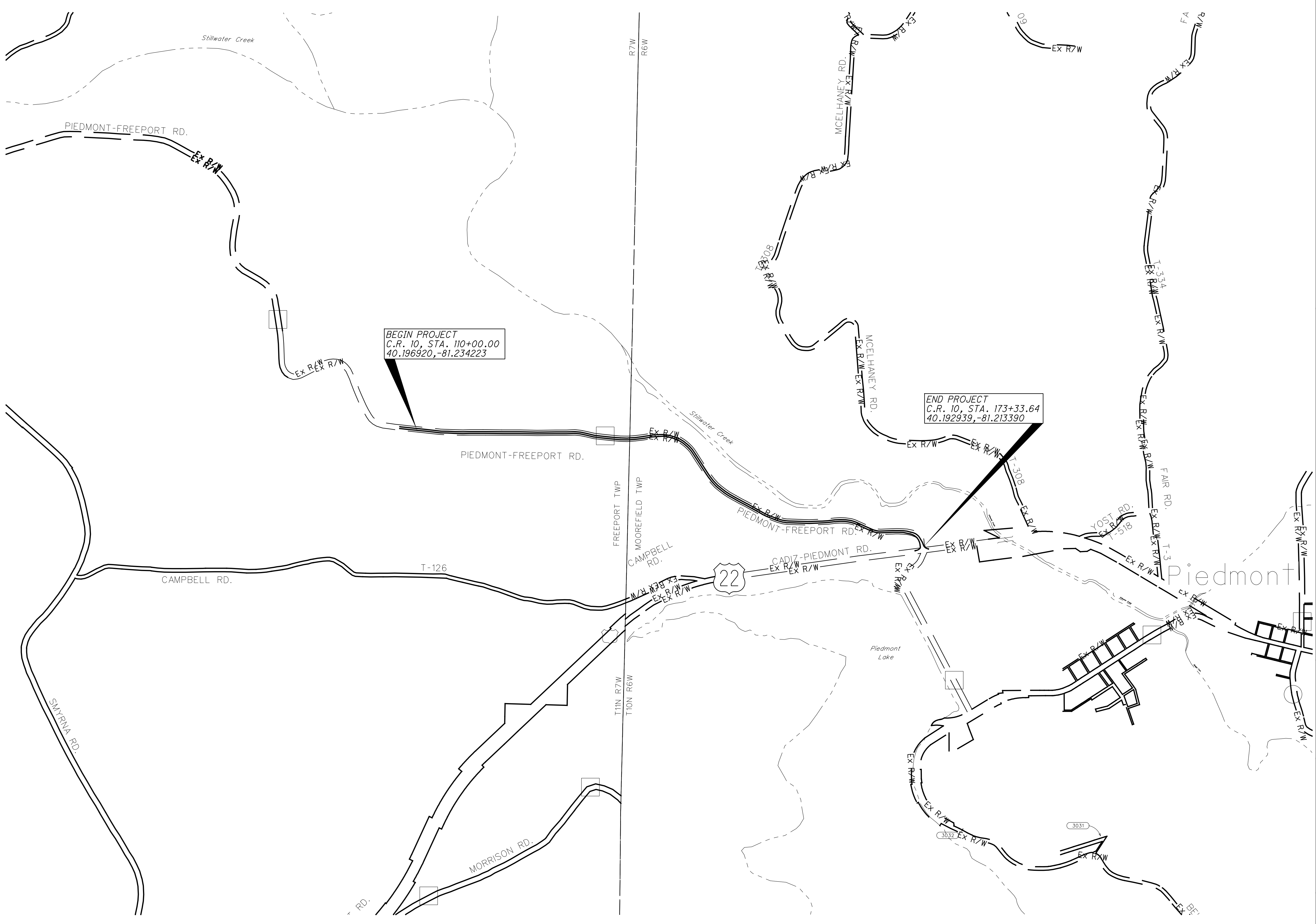

BOUSKA WELL
HAUL ROUTE MAP
PIEDMONT-FREEPORT RD. (C.R. 10)
FREEPORT & MOOREFIELD TWP., HARRISON CO., OH

REV. NO.	SHEET NO.	DATE	DESCRIPTION

PLANS PREPARED FOR:
AMERICAN ENERGY PARTNERS

0 250 500 1000
HORIZONTAL SCALE IN FEET

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